

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1014	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	R.SHOBANA SHRI	Reg. No.:	1921260053
Date:	06.10.2026	Duration:	60 minutes

Code of Conduct

I R.SHOBANA SHRI_(1921260053) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R.shobanashri

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. Understand the Problem Statement

- Explain the problem in your own words.
- Identify the objectives of the proposed system.
- State the scope of the problem.

2. Identify Key Stakeholders and Challenges

- Identify all the stakeholders involved in the system.
- Explain the roles and responsibilities of each stakeholder.
- Discuss the major challenges and issues faced by the stakeholders.

3. Analyze the Current Approach

- Describe the existing system or current process.
- Identify its strengths and limitations.
- Analyze the problems or gaps that need to be addressed.

4. Propose a Suitable Solution Using Software Engineering Principles

- Design a software-based solution for the given problem.
- Describe the major functional modules of the proposed system.
- Identify the functional and non-functional requirements.
- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. Justify the Chosen Methodology/Framework

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).

- Justify why the selected methodology/framework is suitable for the proposed system.
 - Explain how it supports successful project development and maintenance.
6. **Discuss Expected Outcomes and Limitations**
- Describe the expected benefits of the proposed solution.
 - Explain how the solution addresses the identified challenges.
 - Discuss potential risks, implementation challenges, and limitations.
 - Suggest possible future enhancements.

Intelligent Digital Museum Collection Loan Management Platform

Industry Context

Museums frequently lend artifacts to exhibitions and partner institutions. Tracking loans, transportation, insurance, condition, and return dates is complex.

Problem Statement

Design and develop a platform that manages artifact loan requests, approvals, transportation, condition assessments, loan periods, and returns.

Objectives

- Manage loan requests.
- Track approval workflows.
- Record artifact conditions.
- Monitor transportation.
- Track loan periods.
- Manage returns.

Expected Outcomes

- Improved artifact loan visibility.
- Reduced administrative effort.
- Better condition tracking.
- Improved inter-institutional coordination.
- Complete loan histories.

Answer:

1. Problem Understanding

Problem:

Museums lend artifacts to other museums and exhibitions. Currently, loan requests, approvals, transportation, insurance, condition, and return dates may be tracked using separate/manual records. This can cause delays, missing information, and difficulty in tracking artifact history.

Objectives:

- Manage artifact loan requests.
- Track approval workflow.
- Record artifact condition.
- Monitor transportation.
- Track loan periods and return dates.
- Maintain complete loan history.

Scope:

The system covers the complete loan lifecycle from **request** → **approval** → **transportation** → **loan period** → **return** → **condition assessment**.

2. Stakeholder & Challenge Analysis

Stakeholder	Role	Challenge
Museum Admin	Manages loans and approvals	Manual documentation
Lending Museum	Provides artifact	Tracking artifact status
Borrowing Institution	Requests artifact	Approval & coordination delays
Transport Team	Transports artifact	Tracking movement safely
Insurance Officer	Handles insurance	Maintaining insurance details
Exhibition Manager	Uses artifact for exhibition	Monitoring loan period

Major Challenges:

- Manual record maintenance
 - Approval delays
 - Difficulty tracking transportation
 - Condition information may be missed
 - Return dates can be overlooked
 - Difficult to access previous loan history
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3. Current System Analysis**Existing System:**

- Loan details may be maintained using paper records, spreadsheets, emails, and separate documents.

- Different departments may maintain different information.

Strengths:

- Simple to start with.
- Existing staff are familiar with the process.
- No complex software required.

Weaknesses:

- Time-consuming manual work.
- Data duplication.
- Difficult to track real-time status.
- Higher chance of human errors.
- Difficult to maintain complete artifact history.

Gap Identified:

There is a need for a **centralized digital system** that provides complete visibility of the artifact loan process.

4. Proposed Software Solution & SE Principles

Proposed Solution

We propose an **Intelligent Digital Museum Collection Loan Management Platform** that centralizes the complete artifact loan process.

Main Modules:

1. Artifact Management
2. Loan Request Management
3. Approval Management
4. Condition Assessment
5. Transportation Tracking
6. Loan Period Monitoring
7. Return Management
8. Loan History & Reports

Software Engineering Principles:

- **Modularity** – Separate modules for different functions.
 - **Scalability** – Can support multiple museums and artifacts.
 - **Maintainability** – Easy to update and modify.
 - **Security** – Role-based access and protected data.
 - **Reliability** – Accurate records and status tracking.
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5. Methodology / Framework Justification

Agile Methodology

We can use **Agile SDLC** because the system contains multiple modules and requirements may change based on museum staff feedback.

Why Agile?

- Development can be divided into small modules.
- Each module can be developed and tested separately.
- Regular feedback can be collected from stakeholders.
- Changes can be easily incorporated.
- Continuous testing improves system quality.

Agile Flow

Requirement Analysis → Sprint Planning → Development → Testing → Review → Feedback → Next Sprint

conclusion

“The proposed system provides a centralized and secure platform to manage the complete artifact loan lifecycle, reducing administrative effort and improving tracking, coordination, and artifact history management.”

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25